10/8/25 CO2-AT3 class Test

Name: T. chaitra
Reg No: 192425217

1. Machine Learning vs Deep Learning

* Machine Learning enables computers to learn Patterns from data using algorithms, while Deep learning uses multi layer neural network to automatically learn complex features.

Machine Learning	Deep Learning
* Traditional ML algorithms	Multi layer neural networks
* works with smaller datasets.	usually requires large datasets.
* Mostly manual	Automatically learned
* Less computationally expensive	More computationally expensive
* Effective for structured data	Excellent for images, audio, video and text.

Real world Applications:

ML : Fraud detection, customer churn

DL : Face recognition, Autonomous vehicles

CONCLUSION:

ML depends on human-designed features where as DL automatically learns hierarchical features using multiple neural network.

2. Representation Learning, width and Dep

Representation learning:

Representation learning is the process by which a neural network automatically learns useful features from raw data instead of depending completely on manually designed features.

example: Image → Edges → Shapes → object parts → object

Depth:

Depth is the number of layers in a neural network

→ More layers allow heirarchical feature learning.

→ useful for complex problems.

width:

width is the number of neurons in a layer.

→ More neurons → greater learning capacity.

→ can learn more features simultaneously.

CONCLUSION:

width controls the Number of features learned at each level, while depth controls the number of hierachical levels of representation.

ReLU VS Leaky ReLU VS ELU

1. **ReLU:** $f(x) = \max(0, x)$

Positive input → output of x

Negative input → output is 0

Fast and Simple

Main problem : Dying ReLU

2. **Leaky ReLU:** $f(x) = \begin{cases} \alpha x, & x < 0 \\ x, & x \geq 0 \end{cases}$

Allows a small negative input.

Reduces the dying ReLU problem

computationally efficient

Requires selection of α

3. **ELU:** $f(x) = \begin{cases} \alpha(e^{x} - 1), & x < 0 \\ x, & x \geq 0 \end{cases}$

Smooth negative Region

Reduces dead neuron

Can improve optimization

More computationally expensive.

USE: ReLU is a common default, LReLU is useful when dying neurons are a concern, and ELU is useful when smoother negative behavior is desired.

4. Unsupervised Learning, RBM and Autoencoder

Unsupervised Learning:

Unsupervised Learning learns patterns from unlabelled data.

unlabelled data

↓

Neural Network

↓

Hidden patterns

Applications include clustering, dimensionality reduction and anomaly detection.

Restricted Boltzmann Machine:

An RBM is a generative neural network consisting of
→ Visible layer
→ Hidden layer

There are no connections between neuron within the same layer.

Auto encoder:

An encoder consists of

$$\boxed{L = \|x - \hat{x}\|^2}$$

Input → encoder → Latent Representation → Decoder → output.

It learns a compact Representation and reconstructs the original Input.